

PROFESSIONAL INSTALLATION MANUAL

Enterprise Server Deployment: Ubuntu Server 26.04 LTS

Prepared for: TechLine Solutions

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Internal Use — IT Department

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1. Introduction

This manual documents the deployment process for an Ubuntu Server 26.04 LTS virtual machine intended to support file sharing, remote administration, database hosting, web hosting, and internal services for TechLine Solutions.

The instructions below are written so that any system administrator, regardless of prior exposure to this specific project, can reproduce the deployment without additional guidance. It covers the installation itself, initial configuration steps, verification checks to confirm the server is working correctly, and a troubleshooting section covering the issues most likely to come up during a similar deployment.

2. Ubuntu Server Installation Guide

2.1 Prerequisites

Before starting, make sure you have:

- A hypervisor installed (Oracle VirtualBox or VMware Workstation)
- The Ubuntu Server 26.04 LTS ISO downloaded from the official Ubuntu website
- At least 40GB of free disk space and 4GB of RAM available on the host machine

2.2 Virtual Machine Specifications

Create a new virtual machine with the following settings:

Component	Specification
Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 virtual processors
Storage	40 GB (VDI or VMDK)
Network	NAT
Optical Drive	Ubuntu Server ISO

2.3 Installation Steps

1. **Boot the VM** from the Ubuntu Server ISO.
2. **Select language:** English.
3. **Select keyboard layout:** English (US), or whichever layout matches your region.
4. **Choose installation type:** Ubuntu Server (standard).
5. **Configure networking:** Accept the DHCP-assigned IP address unless a static IP is required. Note the assigned address for later use.
6. **Set the hostname:** server01
7. **Create the administrative user account.** Use a non-root account with a strong password. Do not use the root account for daily administration.

8. **Configure disk partitioning:** Choose "Guided – Use Entire Disk." Enable LVM if prompted, and use the ext4 file system. This gives the system flexibility to resize partitions later without repartitioning the whole disk.
9. **Enable SSH:** When prompted, select "Install OpenSSH Server." This is required for remote administration.
10. **Skip additional server snaps** unless the deployment specifically calls for them. Installing extra packages during setup increases the attack surface unnecessarily.
11. **Complete the installation,** remove the installation media when prompted, and reboot.

3. Initial Server Configuration

Once the server has rebooted and you've logged in with the account created during installation, complete the following setup steps.

3.1 Update the System

Keeping the system current is the first thing to do on any new server.

```
sudo apt update
sudo apt upgrade -y
```

3.2 Confirm Network Settings

Check that the assigned IP address matches what was recorded during installation:

```
ip addr
```

3.3 Confirm SSH Is Running

SSH is the primary access point for remote administration, so confirm it's active before relying on it:

```
systemctl status ssh
```

The output should show active (running).

3.4 Set the Hostname (if not already set)

If the hostname needs to be changed after installation:

```
sudo hostnamectl set-hostname server01
```

4. Server Verification Procedures

Use the checklist below to confirm the server is fully operational before handing it off or putting it into use.

Check	Command	Expected Result
User login	N/A	Successful login with created account
Hostname	hostname	Returns server01
IP address	ip addr	Shows assigned IP on the correct interface
Internet connectivity	ping -c 4 google.com	4 successful replies, 0% packet loss

System updates	<code>sudo apt update && sudo apt upgrade -y</code>	Completes without errors
SSH service	<code>systemctl status ssh</code>	Shows active (running)

If any of these checks fail, refer to the troubleshooting section below before proceeding further.

5. Troubleshooting Guide

VM crashes or shows a critical error during installation This usually happens when the installer's on-screen navigation is done with arrow keys instead of Tab. Restart the VM and use Tab to move between fields during setup. If the crash happens repeatedly, confirm the host machine has enough free RAM and CPU allocated to the VM.

apt upgrade fails with a 403 Forbidden error on a specific package This is often caused by a regional mirror that's missing a package or temporarily unavailable. Switch to the global Ubuntu archive mirror:

```
sudo sed -i
's|http://ph.archive.ubuntu.com/ubuntu|http://archive.ubuntu.com/ubuntu|g' /etc/apt/sources.list
```

Then rerun the update and upgrade commands. Before assuming this is necessary, check whether the failing package is actually required for the server's role. Firmware packages for hardware that doesn't exist in a VM (like GPU firmware) are usually safe to skip.

SSH service is not running Confirm it was selected during installation. If it wasn't installed, add it manually:

```
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh
```

No internet connectivity inside the VM Check that the VM's network adapter is set to NAT or Bridged in the hypervisor settings, not "Not attached." Restart the network service if needed:

```
sudo systemctl restart systemd-networkd
```

Cannot find the correct installation ISO Always download from the official source (ubuntu.com for Ubuntu Server, or the Microsoft Evaluation Center for Windows Server). Third-party mirrors or older download pages can lead to outdated or mismatched ISOs.

6. Best Practices

- Never use the root account for routine administration. Create a dedicated user with sudo privileges instead.
- Update the system immediately after installation, and keep a regular update schedule afterward.
- Only install the packages and services actually needed for the server's role. Fewer installed packages means a smaller attack surface.

- Document the assigned IP address, hostname, and any configuration changes as they happen, not after the fact.
- Confirm SSH access works before relying on it for remote management, and disable password authentication in favor of SSH keys once initial setup is complete, if the environment allows it.
- Keep installation media and ISOs organized and verified against official checksums before use.

7. References

- Ubuntu Server Documentation. <https://ubuntu.com/server/docs>
- Canonical Ubuntu Server Installation Guide.
<https://ubuntu.com/server/docs/install/step-by-step>
- OpenSSH Documentation. <https://www.openssh.com/manual.html>